

Data Preprocessing and Quality Assurance Report

1. Introduction

This document outlines the preprocessing steps and quality assurance (QA) protocols applied to ensure dataset reliability, consistency, and suitability for YOLOv26 training.

2. Image Preprocessing

All images were standardized through:

- Automatic resizing to 640×640 pixels
 - Pixel normalization handled internally by YOLOv26
 - Consistent aspect ratio preservation via padding
-

3. Data Augmentation

Roboflow augmentation techniques were applied to improve robustness:

- Horizontal flipping
- Random scaling
- Color jitter (brightness/contrast)
- Minor rotation

Augmentation increased dataset variability without altering semantic labels.

4. Dataset Size Sufficiency

The final dataset meets YOLO modeling requirements:

- 1,000 total object instances
- Multiple instances per class
- Adequate samples for major classes (SUV, Car)

Minority classes (Bicycle, Motorcycle) remain underrepresented and are discussed in model evaluation.

5. Noise Reduction

Quality control included:

- Removal of corrupted or blurry images
 - Re-annotation of ambiguous bounding boxes
 - Background-only images excluded
-

6. Motion Stabilization

Since the dataset consists of static images (not raw video frames), motion stabilization was not required. However, frame selection avoided motion blur.

7. Inter-Annotator Agreement

Annotations were performed by a single annotator following strict labeling guidelines, ensuring internal consistency across the dataset.

8. Class Imbalance Analysis

Class distribution analysis revealed:

- SUV dominance (~58% of validation instances)
 - Severe underrepresentation of Bicycle and Motorcycle classes
-

9. Balancing Strategies (Applied & Proposed)

Applied:

- Data augmentation for minority classes

Proposed:

- Targeted data collection for bicycles and motorcycles
 - Class-weighted loss functions
 - Oversampling minority-class images
-

10. Conclusion

The dataset passed QA checks for structure, labeling consistency, and training readiness. Class imbalance remains the primary limitation influencing detection performance.